

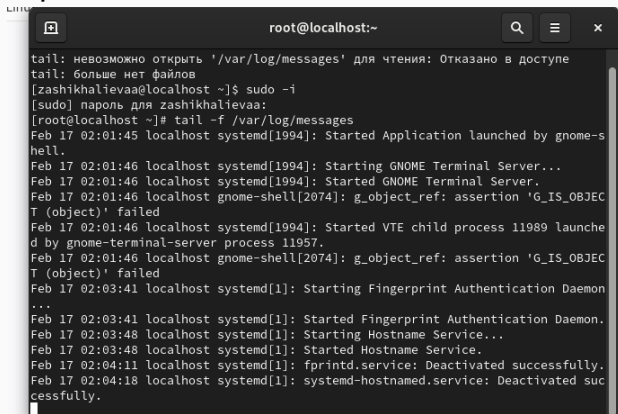
Лабораторная работа№7

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Цель данной лабораторной работы – Получить навыки работы с журналами мониторинга различных событий в системе.

Мониторинг журнала системных событий в реальном времени



```
root@localhost:~  
tail: невозможно открыть '/var/log/messages' для чтения: Отказано в доступе  
tail: больше нет файлов  
[zashikhalievaa@localhost ~]$ sudo -i  
[sudo] пароль для zashikhalievaa:  
[root@localhost ~]# tail -f /var/log/messages  
Feb 17 02:01:45 localhost systemd[1994]: Started Application launched by gnome-shell.  
Feb 17 02:01:46 localhost systemd[1994]: Starting GNOME Terminal Server...  
Feb 17 02:01:46 localhost systemd[1994]: Started GNOME Terminal Server.  
Feb 17 02:01:46 localhost gnome-shell[2074]: g_object_ref: assertion 'G_IS_OBJECT (object)' failed  
Feb 17 02:01:46 localhost systemd[1994]: Started VTE child process 11989 launched by gnome-terminal-server process 11957.  
Feb 17 02:01:46 localhost gnome-shell[2074]: g_object_ref: assertion 'G_IS_OBJECT (object)' failed  
Feb 17 02:03:41 localhost systemd[1]: Starting Fingerprint Authentication Daemon  
...  
Feb 17 02:03:41 localhost systemd[1]: Started Fingerprint Authentication Daemon.  
Feb 17 02:03:48 localhost systemd[1]: Starting Hostname Service...  
Feb 17 02:03:48 localhost systemd[1]: Started Hostname Service.  
Feb 17 02:04:11 localhost systemd[1]: fprintd.service: Deactivated successfully.  
Feb 17 02:04:18 localhost systemd[1]: systemd-hostnamed.service: Deactivated successfully.
```

Изменение правил rsyslog.conf

```
^C
[root@localhost ~]# tail -n 20 /var/log/secure
Feb 17 02:01:07 localhost gdm-password[11870]: gkr-pam: unlocked login keyring
Feb 17 02:01:37 localhost su[10842]: pam_unix(su-l:session): session closed for user alice
Feb 17 02:01:37 localhost su[10652]: pam_unix(su-l:session): session closed for user bob
Feb 17 02:01:37 localhost sudo[10258]: pam_unix(sudo-i:session): session closed for user root
Feb 17 02:01:37 localhost sudo[10840]: pam_unix(sudo:session): session closed for user root
Feb 17 02:01:37 localhost sudo[10650]: pam_unix(sudo:session): session closed for user root
Feb 17 02:03:48 localhost sudo[12052]: zashikhalievaa : TTY=pts/0 ; PWD=/root ; USER=root ; COMMAND=/bin/bash
Feb 17 02:03:48 localhost sudo[12052]: pam_unix(sudo-i:session): session opened for user root(uid=0) by zashikhalievaa(uid=1002)
Feb 17 02:07:22 localhost sudo[12205]: zashikhalievaa : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/bin/bash
Feb 17 02:07:22 localhost sudo[12205]: pam_unix(sudo-i:session): session opened for user root(uid=0) by zashikhalievaa(uid=1002)
Feb 17 02:07:28 localhost sudo[12205]: pam_unix(sudo-i:session): session closed for user root
Feb 17 02:07:54 localhost sudo[12261]: zashikhalievaa : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/bin/bash
Feb 17 02:07:54 localhost sudo[12261]: pam_unix(sudo-i:session): session opened for user root(uid=0) by zashikhalievaa(uid=1002)
Feb 17 02:08:07 localhost sudo[12294]: root : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/bin/bash
Feb 17 02:08:07 localhost sudo[12294]: pam_unix(sudo-i:session): session opened for user root(uid=0) by zashikhalievaa(uid=0)
Feb 17 02:08:30 localhost sudo[12294]: pam_unix(sudo-i:session): session closed for user root
Feb 17 02:08:31 localhost sudo[12261]: pam_unix(sudo-i:session): session closed for user root
Feb 17 02:08:38 localhost sudo[12335]: zashikhalievaa : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/bin/bash
Feb 17 02:08:38 localhost sudo[12335]: pam_unix(sudo-i:session): session opened for user root(uid=0) by zashikhalievaa(uid=1002)
Feb 17 02:09:31 localhost sudo[12335]: pam_unix(sudo-i:session): session closed for user root
[root@localhost ~]#
```

Использование journalctl

```
[root@localhost ~]# journalctl
jue 16 21:27:33 localhost kernel: Linux version 5.14.0-503.23.2.el9_5.x86_64 (mockbuild@ad1-prod-build001.bld.eur.rockylinux.org) (gcc (GCC) 11.5.0 20240710)
jue 16 21:27:33 localhost kernel: The list of certified hardware and cloud instances for Enterprise Linux 9 can be viewed at the Red Hat Ecosystem Catalog, https://www.redhat.com/en/tech-rhel9
jue 16 21:27:33 localhost kernel: Command line: BOOT_IMAGE=(hdd,sdosi)/vmlinuz-5.14.0-503.23.2.el9_5.x86_64 root=/dev/mapper/rl-root ro crashkernel=10-4G:19G
jue 16 21:27:33 localhost kernel: BIOS-provided physical RAM map:
jue 16 21:27:33 localhost kernel: BIOS-e820: [mem 0x0000000000000000-0x0000000000097bfff] usable
jue 16 21:27:33 localhost kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
jue 16 21:27:33 localhost kernel: BIOS-e820: [mem 0x00000000000f0000-0x000000000009ffff] reserved
jue 16 21:27:33 localhost kernel: BIOS-e820: [mem 0x0000000001000000-0x0000000007ffeffff] usable
jue 16 21:27:33 localhost kernel: BIOS-e820: [mem 0x000000007fff0000-0x000000007ffffffffff] ACPI data
jue 16 21:27:33 localhost kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
jue 16 21:27:33 localhost kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
jue 16 21:27:33 localhost kernel: BIOS-e820: [mem 0x00000000ffff0000-0x00000000ffffffffff] reserved
jue 16 21:27:33 localhost kernel: NX (Execute Disable) protection: active
jue 16 21:27:33 localhost kernel: APIC: Static calls initialized
jue 16 21:27:33 localhost kernel: SMBIOS 2.5 present.
jue 16 21:27:33 localhost kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
jue 16 21:27:33 localhost kernel: Hypervisor detected: KVM
jue 16 21:27:33 localhost kernel: kvm-clock: Using msrc 4b564d01 and 4b564d00
jue 16 21:27:33 localhost kernel: kvm-clock: using sched offset of 486438321 cycles
jue 16 21:27:33 localhost kernel: clocksource: kvm-clock: mask: 0xffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle_ns: 881598591483 ns
jue 16 21:27:33 localhost kernel: tsc: Detected 2496,004 MHz processor
jue 16 21:27:33 localhost kernel: e820: update [mem 0x00000000-0x0000ffff] usable ==> reserved
jue 16 21:27:33 localhost kernel: e820: remove [mem 0x000a0000-0x0000ffff] usable
jue 16 21:27:33 localhost kernel: last_pfn = 0x7ffff0 max_arch_pfn = 0x400000000
jue 16 21:27:33 localhost kernel: MTRRs disabled by BIOS
jue 16 21:27:33 localhost kernel: x86/PAT: Configuration [0-7]: WB UC WC UC- WB MP UC- WT
jue 16 21:27:33 localhost kernel: found SMP MP-table at [mem 0x0009ffff-0x0009ffff]
jue 16 21:27:33 localhost kernel: Incomplete global flushes, disabling PCID
jue 16 21:27:33 localhost kernel: RAMDISK: [mem 0x30f0f000-0x3477ffff]
jue 16 21:27:33 localhost kernel: ACPI: Early table checksum verification disabled
jue 16 21:27:33 localhost kernel: ACPI: RSDP 0x0000000000000000 000024 (v02 VBOX )
jue 16 21:27:33 localhost kernel: ACPI: XSDT 0x000000007ffff0030 00003C (v01 VBOX VBOXXSDT 00000001 ASL 00000061)
jue 16 21:27:33 localhost kernel: ACPI: FACP 0x000000007ffff00f0 0000f4 (v04 VBOX VBOXFACP 00000001 ASL 00000061)
jue 16 21:27:33 localhost kernel: ACPI: DSDT 0x000000007ffff0610 002353 (v02 VBOX VBOXBIOS 00000002 INTL 20100528)
jue 16 21:27:33 localhost kernel: ACPI: FACS 0x000000007ffff0200 000040 (v02 VBOX VBOXFACS 00000001 ASL 00000061)
jue 16 21:27:33 localhost kernel: ACPI: APIC 0x000000007ffff0240 000054 (v02 VBOX VBOXAPIC 00000001 ASL 00000061)
jue 16 21:27:33 localhost kernel: ACPI: SSDT 0x000000007ffff02a0 00036C (v01 VBOX VBOXCPUPT 00000002 INTL 20100528)
jue 16 21:27:33 localhost kernel: ACPI: Reserving FACP table memory at [mem 0x7ffff00f0-0x7ffff01e3]
jue 16 21:27:33 localhost kernel: ACPI: Reserving DSDT table memory at [mem 0x7ffff0610-0x7ffff0262]
jue 16 21:27:33 localhost kernel: ACPI: Reserving FACS table memory at [mem 0x7ffff0200-0x7ffff02f3]
jue 16 21:27:33 localhost kernel: ACPI: Reserving APIC table memory at [mem 0x7ffff0200-0x7ffff023f]
```

Для использования фильтрации просмотра конкретных параметров журнала введите `journalctl` и дважды нажмите клавишу `Tab`

```
[CODELINE=
CODE_FILE=
CODE_FUNC=
CODE_LINE=
_COWM=
CPU_USAGE_NSEC=
CURRENT_USE=
CURRENT_USE_PRETTY=
DBUS_BROKER_LOG_DROPPED=
DBUS_BROKER_METRICS_DISPATCH_AVG=
DBUS_BROKER_METRICS_DISPATCH_COUNT=
DBUS_BROKER_METRICS_DISPATCH_MAX=
DBUS_BROKER_METRICS_DISPATCH_MIN=
DBUS_BROKER_METRICS_DISPATCH_STDDEV=
DISK_AVAILABLE=
DISK_AVAILABLE_PRETTY=
DISK_KEEP_FREE=
DISK_KEEP_FREE_PRETTY=
[roo@localhost ~]$ journalctl
AUDIT_LOGINID=
AUDIT_SESSION=
AVAILABLE=
AVAILABLE_PRETTY=
BOOT_ID=
CAP_EFFECTIVE=
_CHDLN=
CODE_FILE=
CODE_FUNC=
CODE_LINE=
_COWM=
CPU_USAGE_NSEC=
CURRENT_USE=
CURRENT_USE_PRETTY=
DBUS_BROKER_LOG_DROPPED=
DBUS_BROKER_METRICS_DISPATCH_AVG=
DBUS_BROKER_METRICS_DISPATCH_COUNT=
DBUS_BROKER_METRICS_DISPATCH_MAX=
DBUS_BROKER_METRICS_DISPATCH_MIN=
DBUS_BROKER_METRICS_DISPATCH_STDDEV=
DISK_AVAILABLE=
DISK_AVAILABLE_PRETTY=
DISK_KEEP_FREE=
DISK_KEEP_FREE_PRETTY=
[roo@localhost ~]$ journalctl I
INITRD_USEC=
INVOCATION_ID=
JOB_ID=
JOB_RESULT=
JOB_TYPE=
JOURNAL_NAME=
JOURNAL_PATH=
_KERNEL_DEVICE=
_KERNEL_SUBSYSTEM=
_KERNEL_USEC=
LEADER=
LIMIT=
LIMIT_PRETTY=
_MACHINE_ID=
MAX_USE=
MAX_USE_PRETTY=
MESSAGE=
MESSAGE_ID=
ERRNO=
_EXE=
_GID=
GLIB_DOMAIN=
GLIB_OLD_LOG_API=
_HOSTNAME=
_INITRD_USEC=
INVOCATION_ID=
JOB_ID=
JOB_RESULT=
JOB_TYPE=
JOURNAL_NAME=
JOURNAL_PATH=
_KERNEL_DEVICE=
_KERNEL_SUBSYSTEM=
_KERNEL_USEC=
LEADER=
LIMIT=
LIMIT_PRETTY=
_MACHINE_ID=
MAX_USE=
MAX_USE_PRETTY=
MESSAGE=
MESSAGE_ID=
_RUNTIME_SCOPE=
SEAT_ID=
_SELINUX_CONTEXT=
SESSION_ID=
_SOURCE_MONOTONIC_TIMESTAMP=
_SOURCE_REALTIME_TIMESTAMP=
SSSD_DOMAIN=
SSSD_PRG_NAME=
_STREAM_ID=
SYSLOG_FACILITY=
SYSLOG_IDENTIFIER=
SYSLOG_PID=
SYSLOG_RAM=
SYSLOG_TIMESTAMP=
_SYSTEMD_CGROUP=
_SYSTEMD_INVOCATION_ID=
_SYSTEMD_OWNER_UID=
_SYSTEMD_SESSION=
_RUNTIME_SCOPE=
SEAT_ID=
_SELINUX_CONTEXT=
SESSION_ID=
_SOURCE_MONOTONIC_TIMESTAMP=
_SOURCE_REALTIME_TIMESTAMP=
SSSD_DOMAIN=
SSSD_PRG_NAME=
_STREAM_ID=
SYSLOG_FACILITY=
SYSLOG_IDENTIFIER=
SYSLOG_PID=
SYSLOG_RAM=
SYSLOG_TIMESTAMP=
_SYSTEMD_CGROUP=
_SYSTEMD_INVOCATION_ID=
_SYSTEMD_OWNER_UID=
_SYSTEMD_SESSION=
TIMESTAMP_BOOTTIME=
TIMESTAMP_MONOTONIC=
_TRANSPORT=
_UDEV_DEVLINK=
_UDEV_DEVNODE=
_UDEV_SYSNAME=
_UID=
UNIT=
UNIT_RESULT=
USER_ID=
USER_INVOCATION_ID=
USERSPACE_USEC=
USER_UNIT=
WP_OBJECT=
WP_OBJECT_TYPE=
_SYSTEMD_SLICE=
_SYSTEMD_UNIT=
_SYSTEMD_USER_SLICE=
_SYSTEMD_USER_UNIT=
TIMEAD_ID=
TID=
TIMESTAMP_BOOTTIME=
TIMESTAMP_MONOTONIC=
_TRANSPORT=
_UDEV_DEVLINK=
_UDEV_DEVNODE=
_UDEV_SYSNAME=
_UID=
UNIT=
UNIT_RESULT=
USER_ID=
USER_INVOCATION_ID=
USERSPACE_USEC=
USER_UNIT=
WP_OBJECT=
WP_OBJECT_TYPE=
```

Если вам нужна детальная информация, то используйте `journalctl -o verbose`

```
[root@localhost ~]# journalctl --since yesterday -p err
фев 16 21:27:33 localhost kernel: Warning: Deprecated Hardware is detected: x86_64-v2:GenuineIntel:12th Gen Intel(R) Core(TM) i5-12450H will not be maintained
фев 16 21:27:33 localhost systemd[1]: Invalid DMI field header.
фев 16 21:27:33 localhost kernel: Warning: Unmaintained driver is detected: e1000
фев 16 21:27:34 localhost kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported hypervisor.
фев 16 21:27:34 localhost kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
фев 16 21:27:34 localhost kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device to avoid problems.
фев 16 21:27:35 localhost systemd[1]: Invalid DMI field header.
фев 16 21:27:37 localhost alsactl[734]: alsa-lib main.c:1554:(snd_use_case_mgr_open) error: failed to import hw:0 use case configuration -2
фев 16 21:27:37 localhost kernel: Warning: Unmaintained driver is detected: ip_set
фев 16 21:28:03 localhost.localdomain gdm-password[1978]: gkr-pam: unable to locate daemon control file
фев 16 21:28:08 localhost.localdomain gdm-wayland-session[1032]: GLib: Source ID 2 was not found when attempting to remove it
фев 16 21:28:08 localhost.localdomain gdm-launch-environment[876]: GLib-GObject: g_object_unref: assertion 'G_IS_OBJECT (object)' failed
фев 17 01:07:06 localhost.localdomain sudo[10761]: bob : user NOT in sudoers ; TTY=pts/1 ; PWD=/home/bob ; USER=root ; COMMAND=/bin/su alice
фев 17 01:07:26 localhost.localdomain sudo[10779]: bob : user NOT in sudoers ; TTY=pts/1 ; PWD=/home/bob ; USER=root ; COMMAND=su-alice
фев 17 01:40:33 localhost.localdomain sudo[11371]: pam_unix(sudo:auth): conversation failed
фев 17 01:40:33 localhost.localdomain sudo[11371]: pam_unix(sudo:auth): auth could not identify password for [alice]
фев 17 01:40:35 localhost.localdomain sudo[11371]: alice : 1 incorrect password attempt ; TTY=pts/2 ; PWD=/data/main ; USER=root ; COMMAND=/bin/su -carol
фев 17 02:00:39 localhost.localdomain gdm-password[11857]: gkr-pam: the password for the login keyring was invalid.
lines 1-18/18 (END)
```

Если вы хотите показать все сообщения с ошибкой приоритета, которые были зафиксированы со вчерашнего дня, то используйте `journalctl --since yesterday -p err`

```
root@localhost ~]# journalctl -o verbose
journalctl: неверный ключ - «o»
root@localhost ~]# journalctl -o verbose
Jun 2025-02-16 21:27:33.486020 MSK [s=e927e32ade4946159e37f149c27c7ec8;i=1;b=653f8f93607143c29007a2c02cefef2b;m=1713cf;t=62e468faf59c4;x=e16495216e1c94b4]
  _SOURCE_MONOTONIC_TIMESTAMP=0
  _TRANSPORT=kernel
  PRIORITY=5
  SYSLOG_FACILITY=0
  SYSLOG_IDENTIFIER=kernel
  MESSAGE=Linux version 5.14.0-503.23.2.el9_5.x86_64 (mockbuild@iad1-prod-build001.bld.equ.rockylinux.org) (gcc (GCC) 11.5.0 20240719 (Red Hat 11.5.0-5), G
  _BOOT_ID=653f8f93607143c29007a2c02cefef2b
  _MACHINE_ID=fd1db2d5c0c74bffa13d35e97b298403
  _HOSTNAME=localhost
  _RUNTIME_SCOPE=initrd
Jun 2025-02-16 21:27:33.486039 MSK [s=e927e32ade4946159e37f149c27c7ec8;i=2;b=653f8f93607143c29007a2c02cefef2b;m=1713e1;t=62e468faf59d7;x=b3d25b235018e839]
  _SOURCE_MONOTONIC_TIMESTAMP=0
  _TRANSPORT=kernel
  PRIORITY=5
  SYSLOG_FACILITY=0
  SYSLOG_IDENTIFIER=kernel
  _BOOT_ID=653f8f93607143c29007a2c02cefef2b
  _MACHINE_ID=fd1db2d5c0c74bffa13d35e97b298403
  _HOSTNAME=localhost
  _RUNTIME_SCOPE=initrd
Jun 2025-02-16 21:27:33.486044 MSK [s=e927e32ade4946159e37f149c27c7ec8;i=3;b=653f8f93607143c29007a2c02cefef2b;m=1713e7;t=62e468faf59dc;x=5d385241fcecff6c]
  _SOURCE_MONOTONIC_TIMESTAMP=0
  _TRANSPORT=kernel
  SYSLOG_FACILITY=0
  SYSLOG_IDENTIFIER=kernel
  _BOOT_ID=653f8f93607143c29007a2c02cefef2b
  _MACHINE_ID=fd1db2d5c0c74bffa13d35e97b298403
  _HOSTNAME=localhost
  _RUNTIME_SCOPE=initrd
```


Заключение: Получены навыки работы с журналом событий.